

A Catalog of Erasure Operations

Mapping the genesis system’s discrete operations by their fan-in

Alexander Pisani & Claude (Anthropic) · June 2026 · companion note (extensible)

Framing. Erasure-only cartography: map values, note where they line up, assert no mechanism of the genesis system. The organising fact is that **every irreversible operation’s erasure is $\ln(\text{fan-in})$** — the log of how many inputs collapse onto the output. The $\ln$ is the generic Landauer signature; the **fan-in is the content**. So to map an operation is to count its fan-in. Tables lead with the **per-output fan-in** (which carries clean closed forms); the prior-dependent *mean* erasure is a secondary thermodynamic summary, shown only where illuminating. Every operation also carries the same two-legged ledger that held for the merge:

$$\underbrace{H(\text{input} \mid \text{output})}_{\text{system erasure}} + \underbrace{H(\text{output})}_{\text{demon reset}} = H(\text{input}) \quad (\text{chain rule}).$$

Claims graded Fact / Reading / Lead. Verified by `erasure_operations.py`, `rewriting_erasure.py`, `modulo_erasure.py`.

1. The additive batch (proto-integers $0..M$, uniform; here $M = 8$)

operation	forgets	per-output fan-in	erasure
add / merge	the split	$n+1$ (output n)	$\ln(n+1)$
projection $(a,b) \rightarrow a$	b entirely	$M+1$ (flat)	$\ln(M+1) = \ln 9$
reset $a \rightarrow 0$	a entirely	$M+1$ (flat)	$\ln(M+1) = \ln 9$
$\max(a,b)$	smaller + which	$2m+1$ (output m)	$\ln(2m+1)$
$\min(a,b)$	larger + which	$2(M-m)+1$ (output m)	$\ln(2(M-m)+1)$

Flat vs graded. Projection and reset are *flat* — constant fan-in $M+1$, forgetting a whole register at a cost independent of its value (reset is the limit: forget everything, record nothing, so demon reset = 0). Max, min, and the merge are *graded* — fan-in varies with the output (**max**: larger max forgot more; **min**: the mirror; merge: $n+1$). Operations differ only in their fan-in *profile*; the $\ln$ wrapper and the two-legged structure are common to all. (*Counts and the flat/graded split: Fact. “Fan-in is the signature”: Reading.*)

2. The string-rewriting family (conditional erasure; length- L strings, $L = 5$, pattern $B = "11"$)

Find the first occurrence of B , then act. The erasure is **conditional** — gated by the match, so the test and the forgetting are fused. Of the 32 length-5 inputs, 13 contain no B and pass through

unchanged (fan-in 1, zero erasure); the other 19 are erased.

operation	forgets	top per-output fan-in	erasure
truncate (find-cut) A → prefix before first B	suffix from B on	empty prefix $\leftarrow 8 = 2^3$	up to $\ln 8$
delete (find-splice) A → A without first B	B's position	4	up to $\ln 4$
substitute A → A with first B → C	that it was B	4	up to $\ln 4$

Conditional / bimodal cost. Per-input erasure is zero for non-matching inputs and positive for matching ones, so the *expected* cost depends on how often the pattern occurs — a dependence with no analogue in the additive family. **Fan-in ranks the family** by how much it discards: truncate (the whole suffix, exponential collapse — shorter prefix, more free suffix) > delete (only *where* B was) > substitute (only *that* it was B). All totals equal $L \cdot \ln 2 = H(\text{input})$. A Turing-complete discrete substrate (Markov / Post / Thue) with no quantum shadow, and the ledger is unchanged. (*Conditional/bimodal structure, ranking, exponential profile: Fact. “Test and forget fused”: Reading.*)

3. Modulo — the multiplicative gateway (inputs 0..M, M+1 = 12 for its divisors)

$a \rightarrow a \bmod m$ keeps the remainder $r = a \bmod m$ and forgets the quotient $q = a/m$. The fan-in of each residue is the quotient $(M+1)/m$.

m	per-residue fan-in	erasure	note
1	$(M+1)/1 = 12$	$\ln 12$	= reset (all → 0)
2	$(M+1)/2 = 6$	$\ln 6$	m 12, uniform
3	$(M+1)/3 = 4$	$\ln 4$	m 12, uniform
4	$(M+1)/4 = 3$	$\ln 3$	m 12, uniform
6	$(M+1)/6 = 2$	$\ln 2$	m 12, uniform
12	$(M+1)/12 = 1$	0	= identity

(Non-divisor m: fan-in near-uniform, $(M+1)/m$ or $(M+1)/m$.)

The fan-in is a division — the first in the catalog. Every prior fan-in was a count or sum; modulo's is a quotient $(M+1)/m$. It interpolates **reset** ($m=1$, fan-in $M+1$) to **identity** ($m=M+1$, fan-in 1), the one-parameter bridge indexed by m.

The ledger becomes a factorization. For $m \mid (M+1)$: **forgotten** = $\ln((M+1)/m)$, **recorded** = $\ln m$, **sum** = $\ln(M+1)$, with $(M+1)/m$ and m a factor pair of $M+1$. Where the merge split the ledger additively, modulo splits it *multiplicatively* — $\ln(M+1)$ along a factorization. The gateway is visible in the ledger itself.

Reconnects to the Stage 1 channels. $a \bmod m$ is the position on the m-cycle — the succession-channel of girth m. Modulo m is the *full* readout of the m-clock (the Stage 1 probe read only “at origin?”), the quotient being how many full m-cycles fit the range.

Honest gateway flag. The clean factor-pair split occurs *exactly* when $m \mid (M+1)$; the pattern of clean-splitting m across all m is the divisor structure of $M+1$ (a prime $M+1$ admits only $m=1$, $m=M+1$). But reading primality off “which m split cleanly” *imports* the divisor structure — the same circularity Stage 1 flagged. The gateway locates the multiplicative structure (the cross- m divisibility pattern = the channel-bank) and warns that getting primality as an *output*, not via the indexing, is the unsolved problem. The erasure values stay non-circular. (*Divisibility–cleanness equivalence: Fact. Circularity caution: methodological, from Stage 1.*)

4. The two-legged ledger is universal

Across every substrate mapped — additive operations, string rewriting, modulo — system erasure plus demon reset equals the input information exactly. Nothing about an operation alters the *structure* of the ledger; only the multiplicity inside the $1n$. For modulo this multiplicity becomes a factor pair. This is the through-line of the catalog and the discipline that keeps it honest: to map an operation is to count its fan-in. (*Fact — chain rule.*)

5. On the horizon

- **The cross- m divisibility pattern** — modulo opened the gateway; the multiplicative/prime structure lives in the divisibility pattern across all m (the Stage 1 channel-bank), not in any single modulo. The standing problem is to make primality emerge there as an *output* of a structural test, not smuggled in through the indexing (the Stage 1 circularity).
- **Parked beyond that** — whether Goldbach prime pairs are cost-extremal (needs arithmetic phase, which needs the multiplicative structure); *cheaper-to-generate more probable to observe*.

6. Grading and honesty boundary

- **Fact** — the fan-in counts and erasures of every operation mapped; flat/graded and max–min mirror symmetry; conditional/bimodal cost and the truncate > delete > substitute ranking; modulo’s quotient fan-in, reset identity interpolation, and factor-pair ledger; the divisibility–cleanness equivalence; the universal two-legged ledger.
- **Reading** — “fan-in profile is the operation’s signature, $1n$ the common wrapper”; “test and forget fused” in rewriting; modulo as the full readout of the Stage 1 m -channel; the ledger going additive $\rightarrow$ multiplicative as the visible gateway crossing.
- **Lead / open** — the cross- m divisibility pattern as the locus of prime structure; primality-as-output (unsolved, circularity to avoid); Goldbach and the observation-probability lead (parked).

Classical content is Landauer (erasure = log multiplicity), the chain rule of entropy, the standard string-rewriting models, and elementary modular arithmetic. The contribution is only the systematic fan-in mapping of these discrete operations across substrates, the flat/graded and conditional-erasure observations, and the additive $\rightarrow$ multiplicative ledger crossing at modulo. No arithmetic question is claimed resolved, and the genesis system’s mechanism is not asserted — only erasure values, and where they line up.

Originating direction (catalog the discrete operations beyond add/count/store; the conditional-truncation example; the deliberate approach to the modulo gateway) is Alexander Pisani’s; the

fan-in mapping and this compilation were developed jointly with Claude (Anthropic). June 2026.